

REV G – HAND-SOLDER PROTOTYPE – FACTORY: U1 / U2 / U4 / L1

J1 NISSAN LEAF CABLE

F1 1A / 125V

D1 B1800 / 100V

CAR\_12V

CAR\_FUSED

VPMF

GND

CAN\_H

CAN\_L

2.2uF

J1 pin 9: +12 V | pin 3: ground | pin 7: CAN-H | pin 2: CAN-L  
Only use the Nissan 200e-HV200 OVM5 cable. Generic OBD cables differ.

65 V buck with input fuse, reverse blocking and 24 V TVS.  
Transient energy, cold operation and parked draw require physical tests.

[illegible]

GPIO3 / GPIO18 / GPIO19 unused. No USB-presence sensing in Rev G.  
OTA updates need firmware implementation. UART remains the recovery interface.

The figure contains two circuit diagrams. The top diagram shows a resistor labeled R21 with a value of R / 1.5W connected between V\_PWR and BULK\_DAMPED. The bottom diagram shows a resistor labeled R22 with a value of R23 / 0.12 connected between V\_PWR and UV\_SENSE, and a capacitor labeled C15 with a value of 100nF / 50V connected between BULK\_DAMPED and GND.

UV monitor is non-latching. Low battery disables the regulator, including Wi-Fi. Delay and hysteresis prevent rapid restart cycling. See protection calculations.

The schematic diagram illustrates the ESP32-C3-WROOM-G2-N4 module and its connections. The module is represented by a yellow rectangle with pins labeled on its left and bottom edges. The left edge pins include ESP\_EN (2), BAT\_SENSE (18), CAN\_STB (17), BOOT\_I02 (16), CAN\_TX (15), CAN\_RX (14), STATUS\_LED (13), BOOT\_I08 (12), BOOT\_I09 (11), and a GND pin (10). The bottom edge pins include a GND pin (9) and a 3V3 pin (8). Various components are connected to these pins: a 10k resistor (R3) to ESP\_EN, a 10uF/16V capacitor (C9) to 3V3, a 100nF/50V capacitor (C10) to 3V3, a 1uF/25V capacitor (C8) to ESP\_EN, a 10k resistor (R4) to BOOT\_I02, a SW1 switch connected to RESET and GND, a 10k resistor (R5) to BOOT\_I08, a 10k resistor (R6) to BOOT\_I09, a SW2 switch connected to BOOT and GND, a 1M/1k resistor (R7) to VFWR, a 10k resistor (R8) to BAT\_DIV, a 1k resistor (R9) to STATUS\_LED, a green LED (D6) to LED\_A, and a 100nF/50V capacitor (C11) to BAT. The module is labeled U2 and ESP32-C3-WROOM-G2-N4.

CAN TX: GP104 | RX: GP105 | STB: GP101  
 STB is pulled high during reset. Drive low only for a requested action.  
 500 kbit/s, standard 11-bit frames; 2013-2015: Leaf profile to be confirmed.  
 TCAN3404: SHDN pin 5 tied to GND, C13 remains an extra rail bypass.

CAN TX: GPIO4 | RX: GPIO5 | STB: GPIO1  
STB is pulled high during reset. Drive low only for a requested action.  
500 kbit/s, standard 11-bit frames; 2013-2015 Leaf profile to be confirmed  
TCAN3404: SHDN pin 5 tied to GND. C13 remains an extra rail bypass.

Rev G: independent UV monitor controls buck EN; no scheduled Wi-Fi disconnect.  
Voltage measurement is after D1 and includes R26; calibrate on the bench.

Factory fits U1 U2 U4 U5 U6 L1. All other parts hand fitted.  
 Prototype: bench, assembly—process and vehicle validation required.

Engineering review and vehicle tests required before release

## DIY Leaf project

Sheet: /  
File: leaf-heat-v7.kicad\_sch

Title: Leaf Heat – Wi-Fi climate control

|          |                  |
|----------|------------------|
| Size: A2 | Date: 2026-09-13 |
|----------|------------------|

Rev: G / prototype

Id: 1/1